

Telecom Customer Churn Prediction

End-to-End Data Analytics Project Report

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1. Project Overview

This project presents a complete, end-to-end data analytics pipeline for predicting customer churn in a telecom company. The goal was to identify customers at high risk of leaving, understand the key drivers of churn, and generate actionable business recommendations, all supported by an interactive Power BI dashboard.

The project covers every stage of the data analytics lifecycle:

1. Data ingestion and exploratory analysis (Python + Pandas)
2. Data cleaning and feature engineering
3. Relational data export to SQL Server
4. Machine learning model training, tuning, and evaluation
5. Prediction on new (unseen) customer data
6. Business intelligence dashboard (Power BI)

2. Dataset Description

The dataset contains 6,418 customer records across 32 features, covering demographics, service subscriptions, billing details, and customer status. After splitting by status, two subsets were created:

Subset	Records	Purpose
churn_data (Stayed + Churned)	6,007	Model training and evaluation
join_data (Joined)	411	Prediction on new customers

Class distribution in churn_data:

Status	Count	Percentage
Stayed	4,275	71.2%
Churned	1,732	28.8%

The dataset is moderately imbalanced. This was addressed using `class_weight='balanced'` in the Random Forest model and `scale_pos_weight` in XGBoost.

3. Data Cleaning & Feature Engineering

3.1 Missing Value Treatment

Missing values were handled with domain-aware imputation rather than deletion:

Column	Missing Count	Fill Strategy
Value_Deal	3,548	Filled with 'None'
Multiple_Lines	622	Filled with 'No'
Internet_Type & related	1,390	Filled with 'No' or 'None'
Churn_Category / Reason	4,686	Filled with 'Others'

3.2 Feature Engineering

Three new features were created to improve model interpretability and dashboard usability:

- Monthly_Charge_Range: Binned into < 20, 20-50, 50-100, > 100
- Age_Group: Binned into < 20, 20-35, 36-50, > 50
- Tenure_Group: Binned into five groups from < 6 Months to >= 24 Months

Numerical sort columns (AgeGrp_Sorting, TenureGrp_Sorting) were also added to enable correct ordering in Power BI visuals.

SQL Server Integration

Cleaned data was exported to SQL Server (database: churn11) using SQLAlchemy with a trusted Windows connection.

KPI Overview – Customer Distribution

Customer Status	Total Customers	Churn Rate
Stayed	4,275	66.61%
Churned	1,732	26.99%
Joined	411	6.40%

Most customers remain active (Stayed), but churn rate is still significant at ~27%, indicating a meaningful retention problem.

Churn by Contract Type

Contract Type	Total Customers	Churned Customers	Churn Rate
Month-to-Month	3,286	1,529	46.53%
One Year	1,413	156	11.04%
Two Year	1,719	47	2.73%

Contract type is the strongest churn predictor. Month-to-month customers are more likely to churn, Long-term contracts significantly improve retention.

Average Revenue per Customer Status

Status	Average Revenue
Stayed	3,745.06
Churned	1,969.95
Joined	119.91

Churned customers generate significantly lower lifetime revenue than retained customers.

Total Revenue Contribution

Status	Total Revenue
Stayed	16,010,148.27
Churned	3,411,960.58
Joined	49,281.56

Although churned customers are fewer, they represent a major revenue loss, making churn reduction financially critical.

Top High-Churn States

State	Churn Rate
Jammu & Kashmir	57.19%
Assam	38.13%
Jharkhand	34.51%
Chhattisgarh	30.51%
Delhi	29.92%

Churn is not uniform geographically — certain states show significantly higher risk, indicating regional service or market issues.

Churn by Reason

Reason	Count	%
Competitor had better devices	289	16.69%
Competitor made better offer	274	15.82%
Attitude of support person	208	12.01%
Unknown / Don't know	124	7.16%
Better data offers	106	6.12%

The primary churn driver is competition, not technical issues.

- Pricing and device competitiveness are key risks
- Customer service experience also strongly impacts churn

Churn by Age Group

Age Group	Total Customers	Churned Customers	Churn Rate
< 20	117	27	23.08%
20-35	1,786	421	23.57%
36-50	1,786	423	23.68%
> 50	2,729	861	31.55%

Older customers (>50) show the highest churn risk.

Churn by Internet Type

Internet Type	Churn Rate
Fiber Optic	41.10%
Cable	25.72%

Internet Type	Churn Rate
DSL	19.37%
None	7.84%

Fiber optic users churn the most, likely due to:

- higher expectations
- higher pricing sensitivity
- service quality issues

Churn by Tenure (Customer Lifecycle)

- Early months (1–12 months) show unstable churn rates (25%–31%)
- Peak churn occurs around 12, 24, and 35 months
- Indicates critical retention periods in lifecycle

Customers are most vulnerable in early lifecycle stages and contract renewal points.

Highest Risk States:

1. Jammu & Kashmir
2. Uttar Pradesh
3. Tamil Nadu
4. Maharashtra
5. Karnataka

These regions require targeted retention campaigns.

The SQL analysis of the telecom customer dataset reveals that customer churn is a significant business challenge, with approximately 26.99% of customers classified as churned. While the majority of customers remain active (66.61%), the churn rate is still high enough to impact long-term revenue stability. The analysis shows that customer behavior varies strongly across different segments, with contract type, service usage, geography, and customer demographics all playing important roles in churn patterns.

Contract type emerges as the most influential factor affecting churn, where Month-to-Month customers show a very high churn rate (46.53%), compared to significantly lower churn among One-Year (11.04%) and Two-Year contracts (2.73%). From a financial perspective, churned customers contribute substantially lower average revenue (1,969.95) compared to retained customers (3,745.06), resulting in a multi-million revenue loss. Additionally, competitive pressure is identified as the primary churn driver, with customers frequently leaving due to better offers, improved devices, and superior competitor services.

Further analysis highlights clear differences across customer segments. Older customers (above 50 years) show higher churn rates, while fiber optic users are the most likely to churn among internet service types. Geographically, states such as Jammu & Kashmir, Uttar

Pradesh, and Tamil Nadu exhibit the highest churn levels, indicating regional risk concentration. Tenure analysis also shows that customers are most vulnerable in the early stages of their lifecycle and around key renewal periods. Overall, the findings demonstrate that churn is driven by a combination of contractual, competitive, and behavioral factors, highlighting the need for targeted retention strategies focused on high-risk segments.

5. Machine Learning Pipeline

5.1 Methodology

A correct and leakage-free pipeline was implemented: train/test split was performed BEFORE label encoding, ensuring that the test set is never seen during encoder fitting. This is a common mistake in many projects and was handled correctly here.

5.2 Models Trained

Model	Train AUC	Test AUC	Accuracy	Notes
Random Forest	0.970	0.892	84%	Best balanced performance
XGBoost	0.934	0.897	82%	Slightly higher AUC

5.3 Best Hyperparameters

Random Forest (via 5-fold GridSearchCV):

- n_estimators: 200, max_depth: 15
- min_samples_leaf: 5, min_samples_split: 20
- class_weight: balanced

XGBoost (via 5-fold GridSearchCV):

- n_estimators: 100, max_depth: 3, learning_rate: 0.1
- min_child_weight: 10, scale_pos_weight: ~2.47

5.4 Model Evaluation — Random Forest

Class	Precision	Recall	F1-Score	Support
Stayed (0)	0.90	0.87	0.89	855
Churned (1)	0.71	0.77	0.74	347
Weighted Avg	0.85	0.84	0.84	1,202

Confusion Matrix: 747 True Negatives, 108 False Positives, 81 False Negatives, 266 True Positives.

The model correctly identifies 77% of churners (recall = 0.77), which is the most business-critical metric, missing a churner is more costly than a false alarm.

5.5 Top Feature Importances

Rank	Feature	Importance
1	Contract Type	24.95%
2	Total Revenue	11.58%
3	Total Charges	10.92%
4	Total Long Distance Charges	7.93%
5	Monthly Charge	7.51%
6	Internet Type	4.44%

Contract type alone explains ~25% of the model's decisions, confirming that month-to-month contracts are the single strongest predictor of churn.

6. Prediction on New Customers

6.1 Results

Prediction	Count	Percentage
At Risk of Churning (1)	386	93.9%
Likely to Stay (0)	25	6.1%
Total New Customers	411	100%

6.2 Why Is Churn So High Among New Customers?

This is not a model error. Analysis reveals four compounding risk factors:

Risk Factor	New Customers	Old Customers	Impact
Month-to-Month Contract	89%	49%	52% churn rate for this contract type
No Online Security	90%	70%	Low service stickiness
No Premium Support	90%	70%	Weak relationship with company
Lower Monthly Charge	43 avg	65 avg	Fewer services, less commitment

Churn rate by contract type (verified from historical data):

- Month-to-Month: 52.4% churn rate
- One Year: 11.2% churn rate
- Two Year: 2.8% churn rate

New customers have 89% month-to-month contracts vs 49% in the historical base. This structural difference fully explains the high predicted churn rate.

7. Business Recommendations

7.1 Immediate Actions (0-3 Months)

- Offer discounted upgrade to 1-year or 2-year contracts for all 386 at-risk new customers. Based on historical data, this alone could reduce their churn rate from 52% to below 11%.
- Provide 3 months of free Online Security to new customers to increase service stickiness and reduce cancellation likelihood.
- Activate Premium Support for new customers during the first 3 months — this is when the relationship is most fragile.

7.2 Medium-Term Strategy (3-12 Months)

- Target customers with high Churn_Probability scores (> 80%) for personal outreach by the retention team.
- Review pricing for Fiber Optic internet customers — this segment shows elevated churn and high monthly charges.
- Investigate why new customers are predominantly choosing month-to-month plans. Consider restructuring the onboarding offer to make annual contracts the default.

7.3 Estimated Business Impact

If churn among the 386 at-risk new customers is reduced from 94% to 30% through contract upgrades and early engagement:

- Estimated customers retained: ~247 additional customers
- Based on average Total Revenue of ~3,034 per customer, this represents approximately 750,000 in protected lifetime revenue